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20 August 2026

Editors-in-Chief
Linear Algebra and its Applications

Dear Editors-in-Chief,

Please consider my manuscript, “Permanental Dominance for Generalized Matrix Functions of Order Four,” for publication as a regular article in *Linear Algebra and its Applications*.

The manuscript proves the complete order-four case of the general subgroup permanental-dominance conjecture. More precisely, for every subgroup $H \leq S_4$, every irreducible complex character χ of H , and every 4×4 Hermitian positive-semidefinite matrix A , it establishes

$$\frac{d_{\chi}^H(A)}{\chi(1)} \leq \text{per } A.$$

This goes beyond the usual S_4 immanant specialization by treating all thirty-seven irreducible-character cases arising from the eleven conjugacy classes of subgroups of S_4 .

The main conceptual contribution is the treatment of the two non-real A_4 characters. Their inequalities are reduced to polynomial nonnegativity on a cone of 3×3 positive-semidefinite Gram matrices and are proved using a rank-one sum-of-squares identity, an exact interior positive-definiteness certificate, rational Gram certificates, and closure. The remaining thirty-five cases are resolved by general principal-minor, moment, and block-contraction inequalities. A companion Lean formalization provides machine-checked counterparts of the central identities and the complete finite reduction.

The result and its methods lie directly within the algebraic, combinatorial, and geometric aspects of matrix theory covered by the journal. The manuscript is original, has not been published previously, and is not under consideration by another journal. A preprint and the companion formalization are publicly available in the cited repository. No external funding was received, and I have no competing interests to declare.

Thank you for considering the manuscript.

Sincerely,

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